

4 Quantification

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The second consists of parameters estimated internally by matching model moments to their empirical counterparts. I describe the two groups in turn, explain which moments are particularly informative about the internally estimated parameters, and then report the model’s fit.¹

4.1 Parameters Set Outside the Estimation

I first describe the parameters assigned outside the estimation. Table 1 summarizes their values and sources.

Demographics and preferences. A model period is four years. Households enter adult life at age 18, retire at age 66, and live at most through age 85. Survival is certain before retirement and thereafter follows four-year probabilities constructed from the combined male and female survivor counts in the 2023 Social Security period life table. Dependent children remain in the household for an expected 18 years, implying a maturity probability of 2/9 per period. I set risk aversion to $\sigma = 2$.

Housing finance and transaction costs. Following Greaney et al. (2025), I set the annual real return on liquid wealth to $i_{\text{ann}} = 0.02$, annual housing depreciation to $\delta_{\text{ann}} = 0.011$, the proportional selling cost to $\psi^s = 0.06$, and the labor-income tax rate to $\tau_{\text{pay}} = 0.179$. The return is based on the average ten-year real interest rate, depreciation combines BEA residential-structure depreciation with an adjustment for the land share of house values, and the tax rate comes from OECD data. I set the annual property-tax rate to $\tau_{\text{ann}}^p = 0.01$. Mortgage debt cannot exceed a share $\phi = 0.80$ of the value of the house, so a purchase requires at least 20 percent equity. The benchmark imposes no separate payment-to-income limit.

Income process. I impose a deterministic lifecycle profile of household earnings, e_a , normalized to average one over working ages. Within age, log income follows an annual AR(1) process with persistence $\rho_z = 0.960$, which I sample at the model’s four-year frequency and approximate with a five-state Rouwenhorst chain. This persistence is near the upper end of the PSID-based estimates summarized by Sommer and Sullivan (2018). I set the annual innovation standard deviation to $\sigma_z = 0.0645$, below estimates of gross income risk in that literature. I treat this variance as provisional rather than as an externally validated estimate of income risk.² The quantitative results are conditional on this restriction.

¹Working text restored from the July 17 reviewed note. Numerical inputs, parameter estimates, and model fit below retain the July reference values pending the agreed item-by-item update; they are not the September experimental results.

²With innovation risk in the empirical range, households in the lowest persistent-income states cannot always finance the model’s minimum consumption and housing bundle. This conflict may reflect missing insurance mechanisms—the current model has no means-tested safety net, family transfers, or labor-supply margin—but it may also indicate that the hard minimum requirements implied by the Stone–Geary specification are too restrictive. Future revisions will therefore reconsider both the missing insurance margins and the Stone–Geary preference structure.

Housing products and supply. Housing is indivisible and available in a limited assortment of types and sizes (Davis and Van Nieuwerburgh, 2015). I represent this lumpiness with five owner-occupied house sizes, $\{H_k\}_{k=1}^5 = \{2, 4, 6, 8, 10\}$ rooms. Renters choose housing continuously up to $\bar{h}^R = 6$ rooms, the 90th percentile of the rental-room distribution in the matched ACS sample. These menus summarize the observed tenure-specific distribution of rooms. I set the housing-supply elasticity to $\eta = 1.75$, following the population-weighted metropolitan estimate in Saiz (2010).

Entry wealth and bequests. Households enter at age 18 with a draw from the PSID net-worth-to-income distribution of childless renters aged 18–24. I impose $\theta_n = 0$, so bequest utility does not vary directly with the number of children. Among PSID households aged 65–75, the difference in median total-wealth-to-income ratios between households with two or more children and one-child households is small and statistically insignificant. Children affect estate accumulation through their effects on housing needs, tenure, and saving.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or restriction
Model period	4 years	Model timing
Entry, retirement, and maximum ages	18, 66, and 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security period life table
Expected years with dependent children	18 years	Child-maturity restriction
Risk aversion σ	2	
Annual real return i_{ann}	2.0%	Average 10-year real rate, 1962–2024; Greaney et al. (2025)
Annual housing depreciation δ_{ann}	1.1%	BEA structures depreciation, adjusted for the residential land share; Greaney et al. (2025)
Annual property tax τ_{ann}^p	1.0%	Maintained tax rate
Financed share ϕ	0.80	20% down-payment restriction; Davis and Van Nieuwerburgh (2015)
Sale cost ψ^s	6%	Davis and Van Nieuwerburgh (2015); Greaney et al. (2025)
Labor-income tax τ_{pay}	17.9%	OECD U.S. average; Greaney et al. (2025)
Income persistence ρ_z	0.960 annually	see text
Income innovation standard deviation σ_z	0.0645 annually	see text
Owner house sizes $\{H_k\}$	$\{2, 4, 6, 8, 10\}$ rooms	Occupied-room distribution in the ACS; see text
Largest rental unit $\bar{h}^R$	6 rooms	Matched ACS rental-size distribution
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Wealth-to-income distribution	Childless renters aged 18–24 in the PSID

4.2 Internally Estimated Parameters

I estimate the remaining 14 parameters internally using 15 moments describing completed fertility, childlessness, housing consumption, tenure, wealth accumulation, and aggregate occupied housing.

Table 2 reports the estimates. Although the parameters are chosen jointly, I organize the discussion into economic blocks and describe which moments are most informative about each block.

Table 2: Internally estimated parameters: July reference

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.9912
α_c	Consumption weight	0.5912
$\bar{c}_0$	Baseline consumption requirement	1.2596
$\bar{c}_n$	Consumption cost per dependent child	0.3929
$\bar{h}_0$	Baseline housing requirement	0.3915
$\bar{h}_{\text{jump}}$	Housing-need increase with the first child	1.6021
$\bar{h}_n$	Housing requirement per dependent child	0.1644
ψ_{child}	Utility from children	0.1979
κ_F	Dispersion in family-size choices	2.0434
χ_O	Utility premium from owner-occupied housing	1.1133
$\bar{H}$	Housing-supply scale	8.6456
θ_0	Strength of the bequest motive	0.3118
θ_1	Bequest curvature shift	0.3973
κ_T	Dispersion in tenure choices	0.0100

Saving. The annual discount factor β_{annual} governs households’ willingness to carry resources across ages. Liquid wealth among young childless renters, the older nonhousing-wealth share, and the late-life estate median are particularly informative about it.

Housing demand, tenure, and supply. I use housing-size differences by family composition to discipline the housing requirements. Mean rooms among childless renters is particularly informative about $\bar{h}_0$, the increase associated with the first child disciplines $\bar{h}_{\text{jump}}$, and the difference between larger and smaller families disciplines $\bar{h}_n$. The owner–renter rooms difference and the share of childless owners in large homes help discipline α_c and χ_O . Ownership rates by age and family status are particularly informative about χ_O and κ_T , while aggregate occupied rooms disciplines the housing-supply scale $\bar{H}$.

Consumption needs and family size. The consumption requirement is $\bar{c}_0 + \bar{c}_n n_s$. Outcomes for childless households, including young liquid wealth, are particularly informative about the baseline requirement $\bar{c}_0$, while fertility and differences in housing and tenure by family composition help discipline the incremental cost $\bar{c}_n$. Completed fertility and childlessness jointly discipline these costs with the direct utility value of children ψ_{child} and the dispersion of family-size choices κ_F . These parameters are estimated jointly because the same family outcomes respond to both the cost and the utility value of children.

Bequests. The bequest parameters θ_0 and θ_1 govern the strength and wealth dependence of the bequest motive. The late-life median estate is particularly informative about θ_0 , while the older nonhousing-wealth share, together with old-age ownership, provides additional information about θ_1 . These moments also respond to saving and tenure choices, so the two bequest parameters are disciplined jointly with those blocks.

4.3 Target Moments and Fit

The target system combines completed fertility and childlessness from the CPS, tenure and housing-size moments by age and family status from the ACS, housing responses to changes in the number of children, and wealth moments from the PSID. Table 3 reports all 15 target moments and their model counterparts.

Table 3: Target moments and model fit: July reference

Moment	Data	Model
<i>Family size and family housing</i>		
Completed fertility	1.918	2.034
Childlessness rate	0.188	0.189
Family ownership gap	0.168	0.219
Housing increment with the first child, rooms	0.664	0.681
Rooms gap, 3+ versus 1–2 children, ages 30–55	0.368	0.607
<i>Tenure and housing consumption</i>		
Ownership rate, ages 30–55	0.575	0.658
Renter mean rooms, no children at home, ages 30–55	3.805	3.963
Owner share with rooms ≥ 6 , no children at home, ages 30–55	0.596	0.760
Owner–renter mean rooms gap, no children at home, ages 30–55	2.419	2.323
Ownership rate, ages 25–34	0.341	0.274
Ownership rate, ages 65–75	0.764	0.954
Mean occupied rooms, ages 18–85	5.780	5.673
<i>Wealth and bequests</i>		
Young childless-renter liquid wealth / annual income, ages 25–35	0.179	0.328
Median total estate wealth / annual income, ages 76–84	6.501	6.431
Share with nonhousing wealth $\geq$ annual income, ages 65–75	0.608	0.610

The model closely matches childlessness, the housing increment associated with the first child, renter mean rooms, the owner–renter rooms difference, aggregate occupied rooms, the late-life median estate, and the older nonhousing-wealth share. It produces completed fertility of 2.034, compared with 1.918 in the data. The model overstates the overall ownership rate, the family ownership gap, the share of childless owners in large homes, and the rooms difference between larger and smaller families, while understating ownership at ages 25–34.

The wealth moments reveal a lifecycle tension. The two late-life wealth targets are closely matched, but young childless renters accumulate liquid wealth equal to 0.328 of annual income, compared with 0.179 in the PSID. The estimated annual discount factor of 0.9912 should therefore be read as a joint calibration outcome rather than as a direct estimate of household patience.

The largest miss is old-age ownership: the model produces an ownership rate of 0.954 at ages 65–75, compared with 0.764 in the ACS. The model has no health or long-term-care expense risk and no age-dependent maintenance burden that induces older owners to leave homeownership. Together with transaction costs and a bequest motive that values the house at death, this makes owner housing unusually persistent late in life. The untargeted estate distribution shows a related limitation: the model produces an estate p90–p50 ratio of 1.75, compared with 3.45 in the PSID, and understates nonhousing wealth in the upper part of the late-life distribution. Part of this gap reflects measurement, because PSID nonhousing wealth includes business wealth, retirement accounts, and other asset classes with no counterpart in the model. I therefore do not use the

current calibration to draw conclusions that depend on the late-life ownership path or portfolio composition.

The calibration remains provisional because the income variance and the late-life exit and portfolio margins require further work. The quantitative exercises that follow focus on family housing needs, tenure, and the down-payment constraint and should be read conditional on these limitations.

References

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